

Manu Mathew Jiss

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

Education

[University of the Pacific](#), Stockton, CA May 2026
Master of Science in Computer Science

[APJ Abdul Kalam Technological University](#), Kerala, India March 2023
Bachelor of Technology in Computer Science and Engineering

Research Interests

Foundation Models, Medical AI, Multimodal AI, Agentic AI, Human AI Collaboration, Large Language Models (LLMs), Computer Vision, Natural Language Processing, Explainable AI, Trustworthy AI, Precision Health, Brain Inspired AI

Technical Skills

Programming Languages:	Python, JavaScript, TypeScript, C++, SQL, Bash/Shell Scripting, MATLAB
Foundation Models & AI:	LLMs, Foundation Models, Hugging Face Transformers, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), Agentic AI, TensorFlow, PyTorch, Scikit-learn, NLP, Explainable AI (XAI)
Computer Vision & Robotics:	CUDA, ROS/ROS2, NVIDIA Jetson, Jetson Orin Nano, ZED Stereo Cameras, LiDAR, IMUs, Sensor Fusion, Raspberry Pi, Embedded Linux, Arduino
Backend Development:	FastAPI, Flask, Fastify, Node.js, Express.js, Java, Spring Boot, REST APIs, OAuth, JWT Authentication, Prisma ORM, Microservices, AWS Lambda
Databases & Data Engineering:	PostgreSQL, MongoDB, Redis, ClickHouse, SQLite, ETL Pipelines, Incremental ETL, Data Warehousing, Database Design, Data Modeling, Metadata Lineage
Cloud & DevOps:	Docker, Docker Compose, GitHub Actions, AWS, Azure, Linux (Ubuntu), CI/CD, Nginx, SSH/SCP
Software Engineering:	Git, GitHub, Object Oriented Programming (OOP), Design Patterns, Data Structures, Algorithms, System Design, Agile Development, Jupyter Notebook
Research Tools & APIs:	OpenAI API, Hugging Face, Google Cloud APIs, Weights & Biases, Reddit API, News-API, Python Research Stack

Graduate Research Assistant – [University of the Pacific](#)

- [Dr. Pramod Gupta](#) December 2025 – May 2026
 - Designed a multimodal emotion inference framework combining RoBERTa, foundation models, and hybrid reasoning to support AI assisted mental health journaling and personalized mood analysis.
 - Developed [AIMoodDiary](#), an AI powered journaling system for evaluating hybrid emotion inference and personalized mood analytics; **paper accepted at IEEE ICMLA 2026**.
- [Dr. Don Lee](#) August 2025 - December 2025
 - Designed a multimodal vision language pipeline integrating CLIP, CUDA, ROS2, and NVIDIA Jetson for robust perception under adversarial conditions.
 - Evaluated the proposed approach against CNN baselines, achieving a 38% improvement in recognition accuracy on manipulated traffic signs; **paper under review at CVPR 2026 (OpenReview)**.
- [Dr. Solomon Berhe](#) May 2025 – May 2026
 - Designed NLP and large language model pipelines for reasoning over software discussions, enabling explainable classification of Technical Problem Solving and General Discontent.

- Proposed an author community sentiment divergence metric and developed an interactive research **dashboard** for model evaluation and analysis; **paper accepted at IEEE ICMLA 2026**.
- *Dr. Tapadhir Das* *December 2024 – May 2025*
 - Designed and evaluated machine learning pipelines for sentiment, toxicity, and account authenticity analysis using XGBoost, feature engineering, and large scale social media datasets.
 - Proposed a risk profiling framework for bot, scam, spam, and real account detection, demonstrating robust classification performance; **paper accepted at IEEE SMC 2026**.
- *Dr. Tapadhir Das* *July 2025 – March 2026*
 - Developed LIMFADD, an LLM assisted dataset generation framework for trustworthy AI research, creating high quality labeled datasets through automated validation.
 - Designed an automated validation pipeline integrating explainability techniques to improve dataset quality and support reproducible account authenticity research; **paper accepted at IEEE ISTAS 2025 (TechRxiv)**.

Software Engineering Experience

- **Software Engineering Intern** May 2023 – July 2024
Equalizer, Dubai, UAE (Remote)
 - Developed distributed, fault tolerant telemetry processing pipelines using Java, Spring Boot, PostgreSQL, Docker, and REST APIs to enable real time infrastructure monitoring, automated failure detection, and high availability for large scale cryptocurrency mining systems.

Startup Experience

- **Co-Founder & Software Engineer** May 2025 – Present
Stock Crusher (GenAI Trading Platform)
 - Designed an agentic AI platform integrating Retrieval Augmented Generation (RAG), multiple foundation models, and multi agent reasoning for explainable decision support.
- **Co-Founder & Software Engineer** October 2025 – Present
Meal Muse (AI Recipe Assistant)
 - Developed production backend services using Fastify, Node.js, TypeScript, PostgreSQL, Prisma ORM, and REST APIs, implementing OAuth authentication, caching, rate limiting, and AI powered recipe generation for a deployed iOS application.

Graduate Teaching Assistant – University of the Pacific

- *Dr. Sethuraman Kuruvimalai* *February 2026 – May 2026*
 - Assisted in teaching cybersecurity analytics, mentoring students in SOC workflows, SIEM, Splunk, Linux environments, virtual machines, and log analysis while supporting assignment design and hands on security investigations.

Selected Publications

- AIMoodDiary: Hybrid Emotion Inference for Reflective Journaling and Mood Analytics. IEEE ICMLA 2026 (Accepted)
- Distinguishing Technical Problem Solving and General Discontent in Reddit Discussions. IEEE ICMLA 2026 (Accepted)
- Template Guided Vision Language Modeling for Robust Traffic Sign Recognition. CVPR 2026 (Under Review)
- Three Phase Pipeline for Social Media Account Authenticity. IEEE SMC 2026 (Accepted)
- LIMFADD: LLM Assisted Fake Account Detection Dataset. IEEE ISTAS 2025 (Published)